ForeSight — Test Case Documentation

Project: AO-02 Capacity Early-Warning Dashboard (Streamlit)

Build tested: current working tree, branch master · **Date:** 29 October 2026

Data: Synthetic demo record (all screenshots). TC-02 also loads the M100 CINECA record, without a screenshot.

15	14	1	6
test cases	passed	failed	features covered

How the tests were run

- **UI tests** (TC-01–TC-11) drive the real `app.py` with Streamlit's `AppTest`: they click widgets and read the rendered page. Screenshots come from the same app served by `streamlit run` and driven in headless Chromium.
- **Pipeline tests** (TC-12–TC-15) call `pipeline.loader` and `pipeline.exhaustion` directly, using small temporary data files. They have no UI, so the evidence shown for them is the test runner's console output.
- The "Actual result" for every case is what the test runner recorded on this run. Nothing was typed in by hand.

Summary

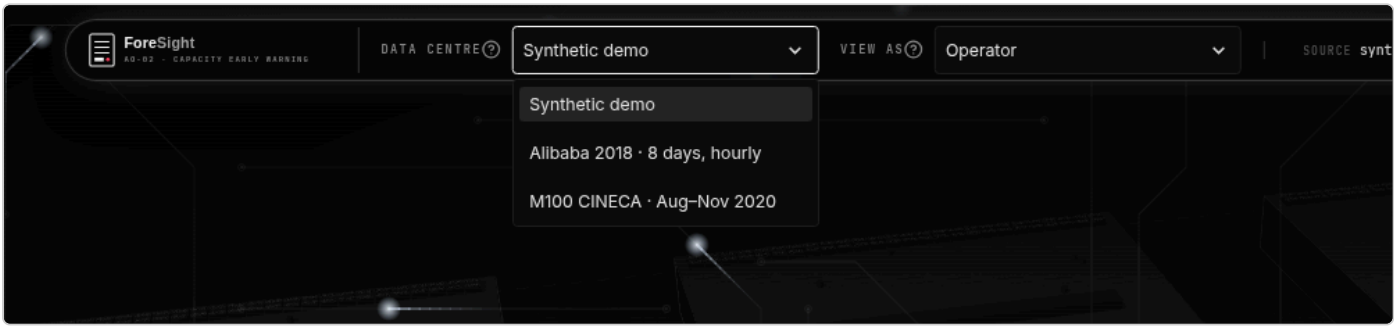
ID	Feature	Type	Status
TC-01	Data centre record selector	Positive	Pass
TC-02	Data centre record selector	Validation	Pass
TC-03	Resources chart (Capacity outlook)	Positive	Pass
TC-04	Resources chart (Capacity outlook)	Negative / boundary	Pass
TC-05	Resources chart: alert lines	Validation	Pass
TC-06	Recommendations tab	Positive	Pass
TC-07	Recommendations: Kind filter	Positive	Pass
TC-08	Recommendations: Combined plan	Negative / boundary	Pass
TC-09	Overview: Recommended plan (Combine toggle)	Validation (required input)	Pass
TC-10	Recommendations: combination maths	Validation	Pass
TC-11	Tenant view (access control)	Validation / security	Pass
TC-12	Data loader: required columns	Negative	Pass
TC-13	Data loader: invalid values	Negative / validation	Pass
TC-14	Forecast ETA (exhaustion)	Boundary	Pass
TC-15	Data loader: invalid timestamp	Negative	Fail

Defect found (TC-15): one malformed timestamp in a telemetry file stops the whole data centre from loading with a `ValueError`. Invalid numeric values are dropped instead (TC-13). Suggested fix: in `pipeline/loader.py` `_normalise()`, pass `errors="coerce"` to `pd.to_datetime`. The row is then dropped by the existing `dropna(subset=["timestamp", ...])`.

TC-01 Data centre record selector

Positive **Pass**

Test Case ID	TC-01
Feature / functionality	Data centre record selector
Test type	Positive
Test input	Open the app with no AO02_DATA_DIR set; open the “Data centre” dropdown in the top bar.
Expected result	Dropdown lists every record (Synthetic demo + each raw_data/dc_* record); “Synthetic demo” is selected by default; page renders with no exception.
Actual result	Selector shows 3 records ['Synthetic demo', 'Alibaba 2018 · 8 days, hourly', 'M100 CINECA · Aug–Nov 2020']; default = 'Synthetic demo'; exceptions = 0
Pass / Fail	**PASS**
Justification	Every other feature depends on loading the right data centre. If the default or the list is wrong, every chart and alert after it is wrong too.



Top bar: Data centre dropdown with all three records

TC-02 Data centre record selector

Validation **Pass**

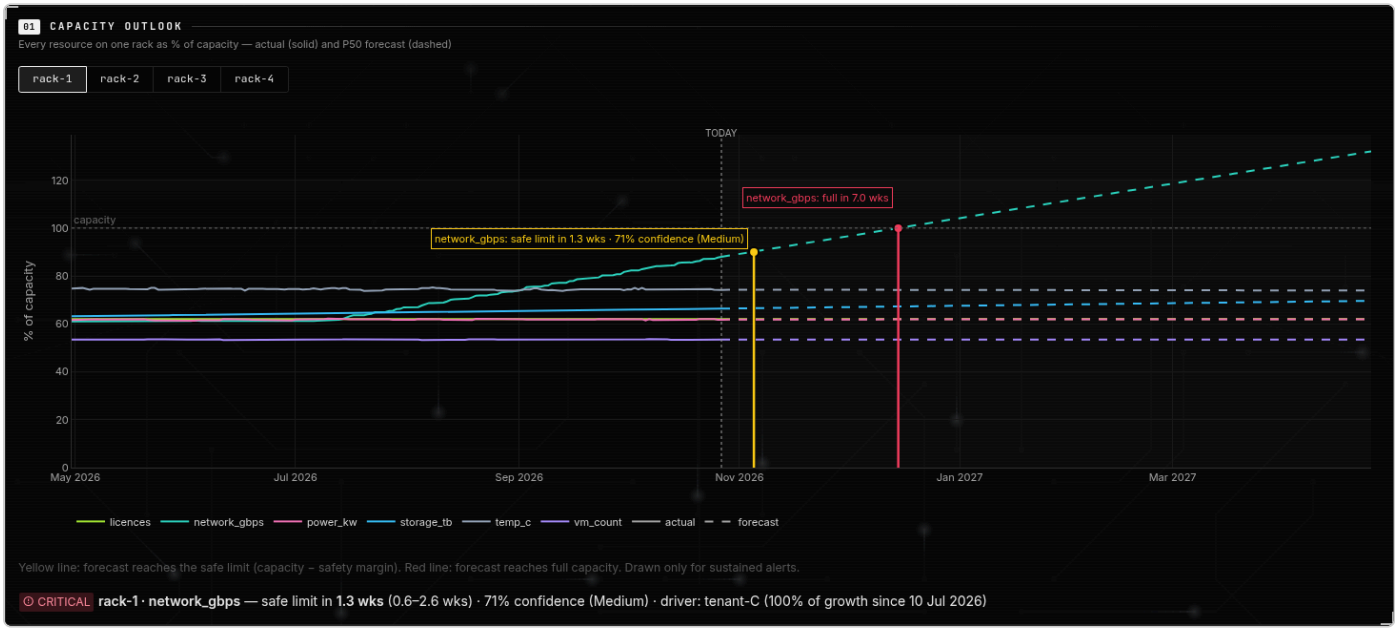
Test Case ID	TC-02
Feature / functionality	Data centre record selector
Test type	Validation
Test input	Pick rack-4 on the chart, then switch the record to “M100 CINECA · Aug–Nov 2020”.
Expected result	App reloads the M100 telemetry with its own racks; the old rack pick (rack-4 of the synthetic record) is not carried over; no exception.
Actual result	Before switch hero rack = 'rack-4'. After selecting 'M100 CINECA · Aug–Nov 2020': no exception (0), racks = 49 racks (rack-00 … rack-48), hero rack reset to 'rack-02' (stale pick not carried over)
Pass / Fail	**PASS**
Justification	Each record has different racks and resources. Stale selections from the previous record could crash the page or show the wrong rack’s data.

```
$ venv/bin/python run_tests.py   # TC-02
TC-02: PASS - Before switch hero rack = 'rack-4'. After selecting 'M100 CINECA · Aug–Nov 2020': no exception (0), racks = 49 racks (rack-00 … rack-48), hero rack reset to 'rack-02' (stale pick not carried over)
```

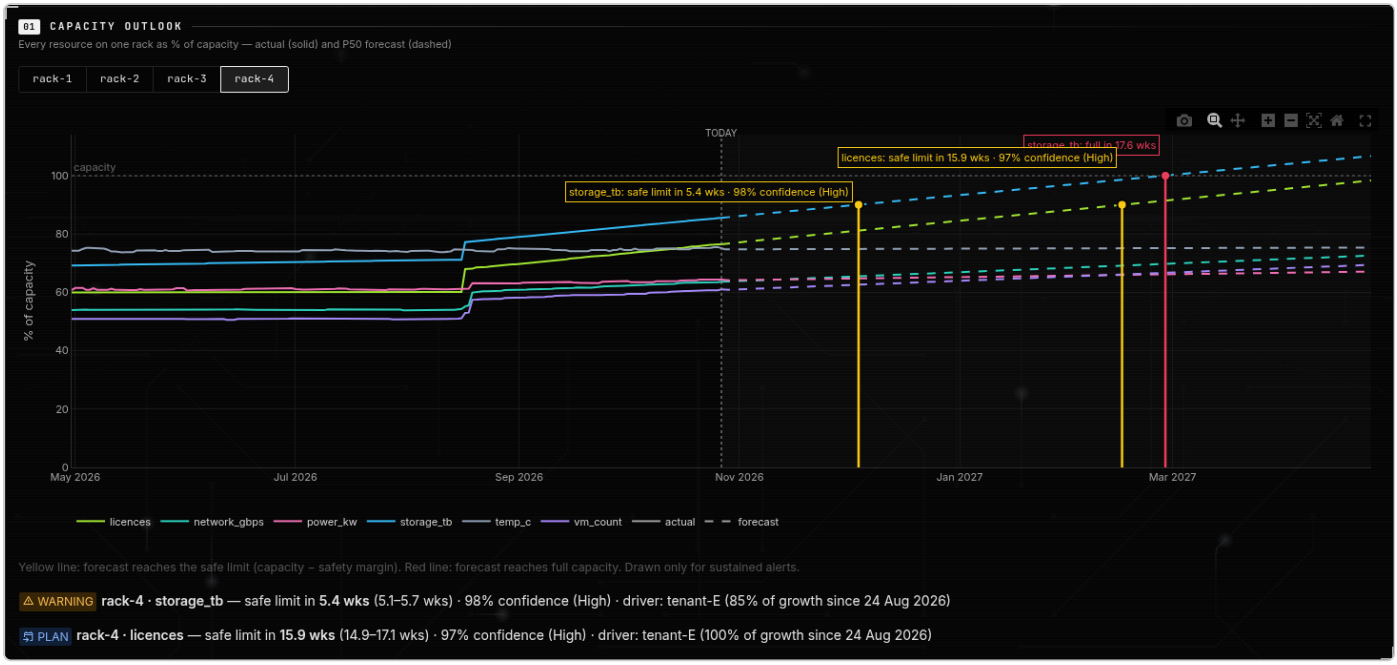
TC-03 Resources chart (Capacity outlook)

Positive | **Pass**

Test Case ID	TC-03
Feature / functionality	Resources chart (Capacity outlook)
Test type	Positive
Test input	Click the “rack-4” button above the Capacity outlook chart.
Expected result	Chart redraws for rack-4: one solid line (actual) and one dashed line (P50 forecast) per resource, as % of capacity, with alert markers for that rack.
Actual result	Racks offered ['rack-1', 'rack-2', 'rack-3', 'rack-4']; clicked 'rack-4'; chart re-rendered for 'rack-4', 19 Plotly charts on page, no exception
Pass / Fail	**PASS**
Justification	The rack selector is the main control on the main chart. This confirms that clicking it actually changes what the chart shows.



Capacity outlook, default (most urgent) rack-1

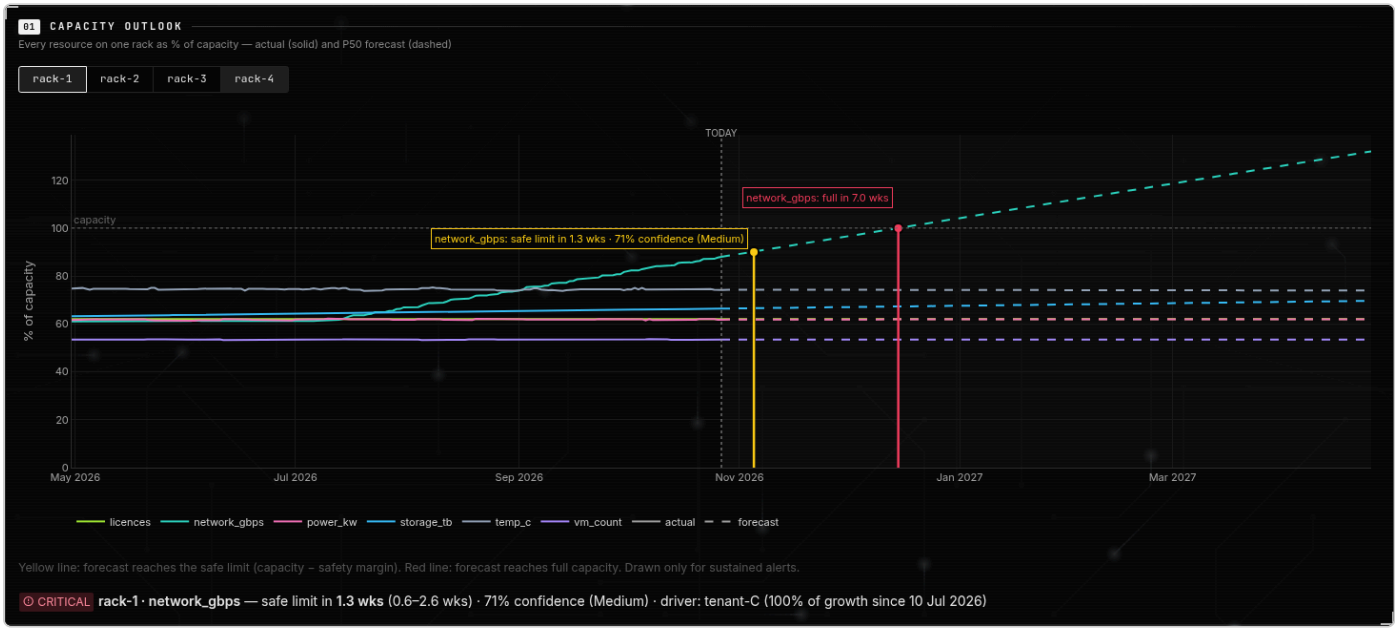


After clicking rack-4

TC-04 Resources chart (Capacity outlook)

Negative / boundary | Pass

Test Case ID	TC-04
Feature / functionality	Resources chart (Capacity outlook)
Test type	Negative / boundary
Test input	Click the already-selected rack button again so the control has no value (None).
Expected result	App does not crash or show an empty chart; it falls back to the worst (most urgent) rack.
Actual result	Rack control cleared (value None). App fell back to worst rack 'rack-1'; 19 charts still rendered; exceptions = 0
Pass / Fail	**PASS**
Justification	Streamlit segmented controls can be deselected, which sends None to the code. An unhandled None here would break the main page.

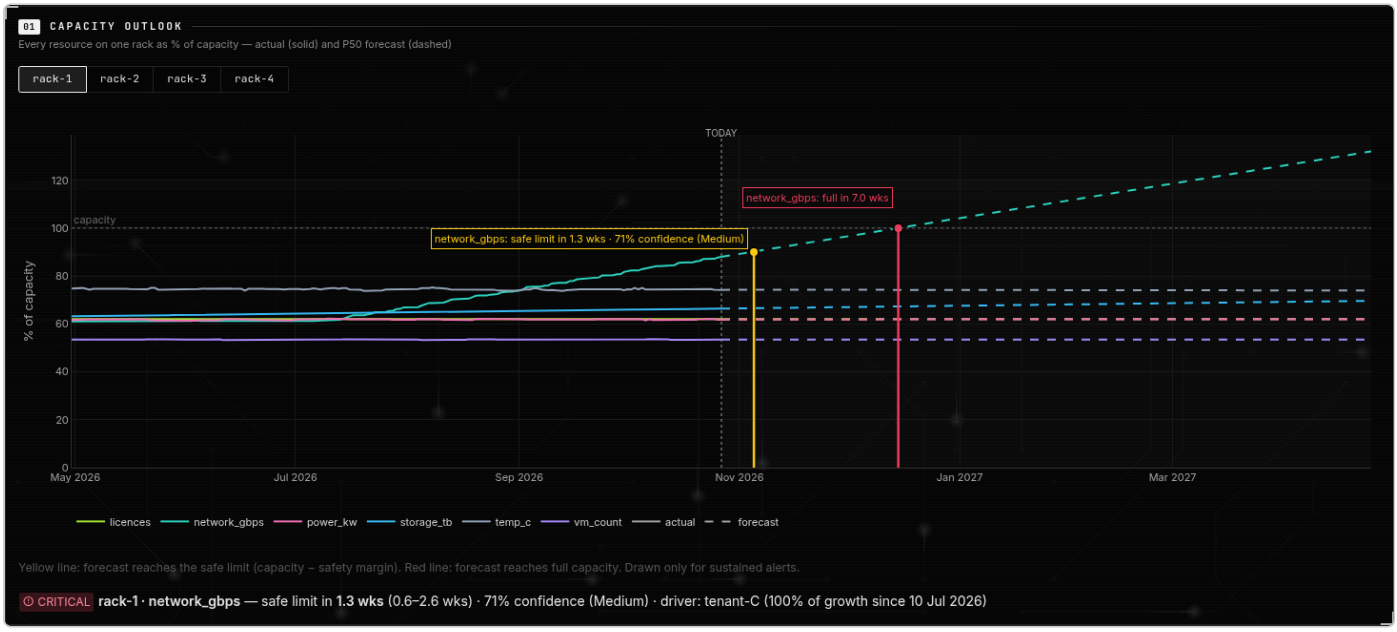


Selected rack clicked again: falls back to the most urgent rack

TC-05 Resources chart: alert lines

Validation | Pass

Test Case ID	TC-05
Feature / functionality	Resources chart: alert lines
Test type	Validation
Test input	Run the forecast on the synthetic record and compare each series' status with the yellow (safe limit) line the chart draws.
Expected result	Every alerting series (CRITICAL/WARNING/PLAN) has a likely ETA that places its yellow line; OK series get no attention line.
Actual result	24 rack×resource series; 4 alerting (rack-1·network_gbps, rack-2·storage_tb, rack-4·licences, rack-4·storage_tb…), each with a likely ETA for the yellow safe-limit line; 20 OK series get no line
Pass / Fail	**PASS**
Justification	The chart's yellow and red lines are what the operator acts on. A line without a matching alert, or an alert without a line, would be misleading.

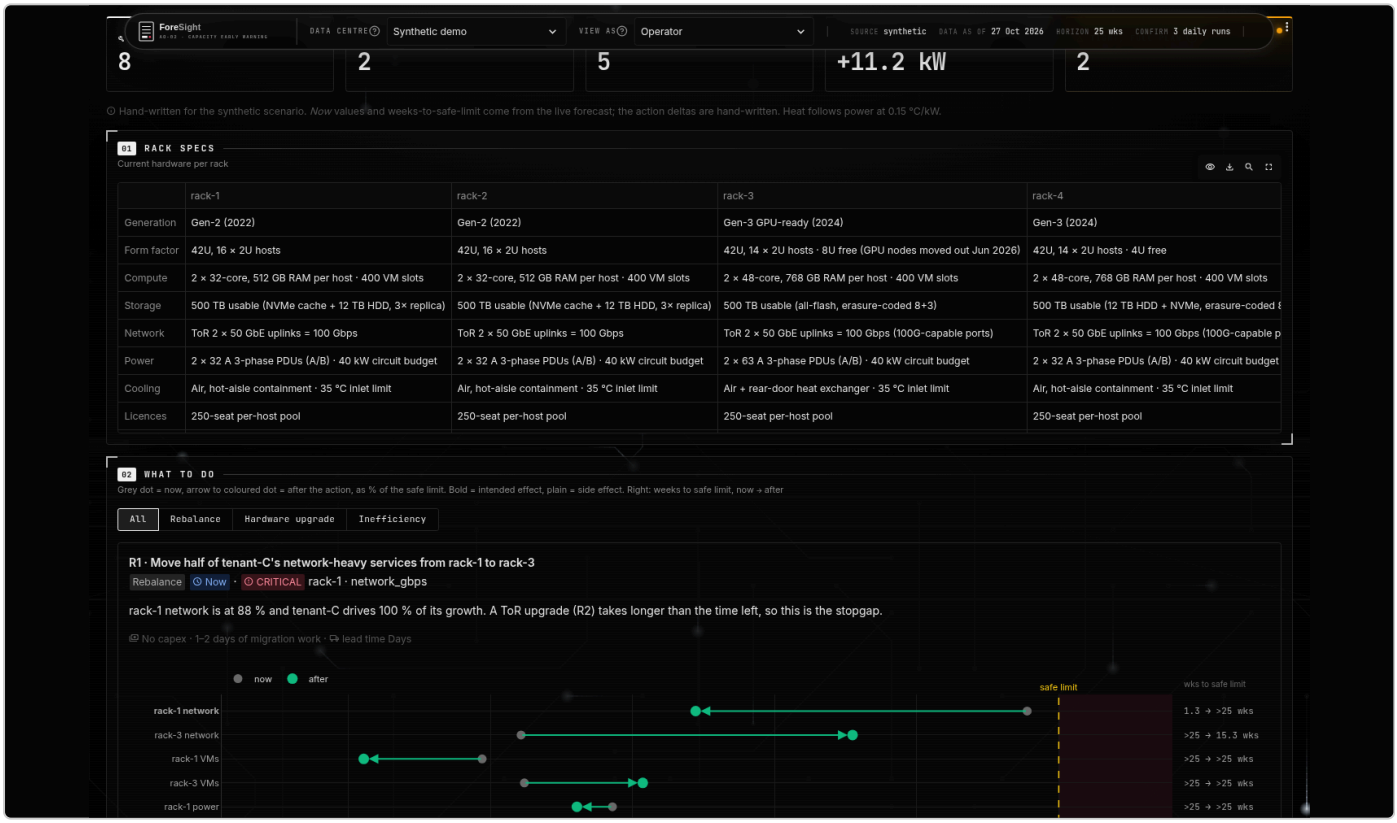


Capacity outlook, default (most urgent) rack-1

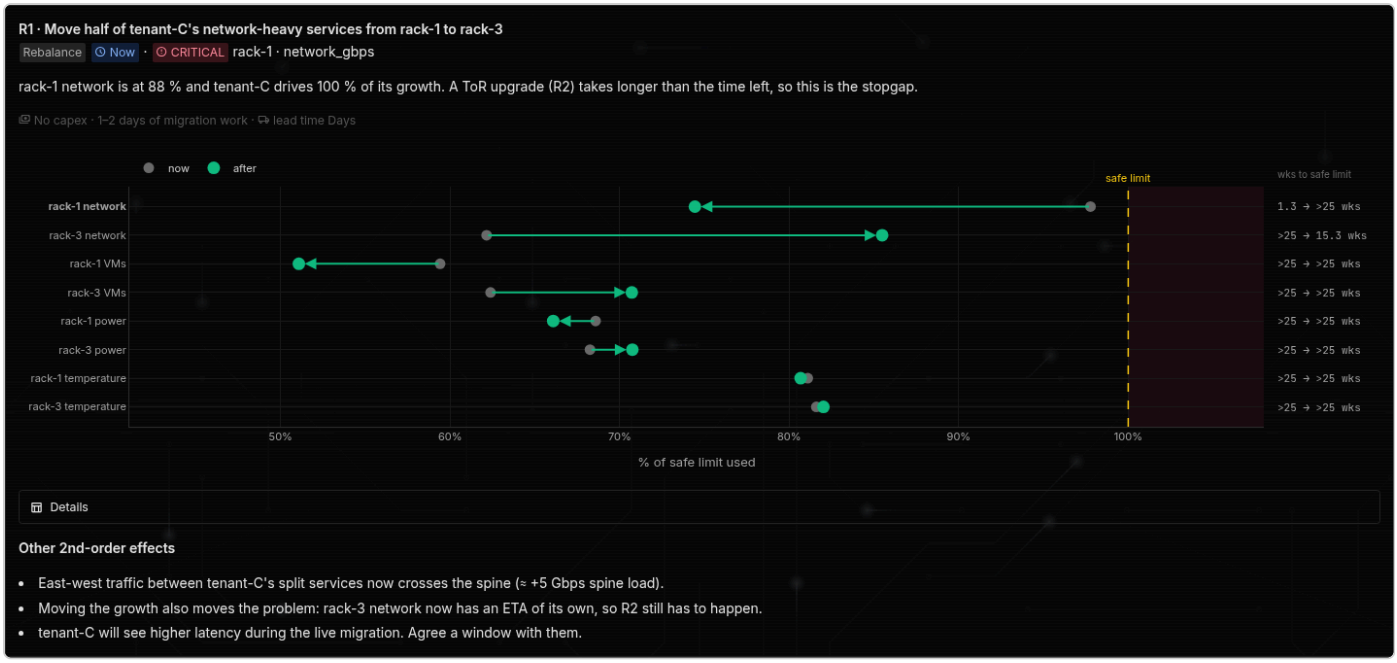
TC-06 Recommendations tab

Positive | Pass

Test Case ID	TC-06
Feature / functionality	Recommendations tab
Test type	Positive
Test input	Open the Recommendations tab (Operator view, synthetic record).
Expected result	KPI row (recommendations, upgrades, inefficiencies, net power, tight rows), one card per action with an effect chart, and a Combined plan panel all render.
Actual result	Recommendations tab rendered 8 actions (Hardware upgrade, Inefficiency, Rebalance), 10 effect charts, 2 'tight' warnings in Combined plan
Pass / Fail	**PASS**
Justification	This is the tab that tells the operator what to do. It must render completely before any of its controls are worth testing.



Recommendations tab: KPI row



A recommendation card with its effect chart

TC-07 Recommendations: Kind filter

Positive | Pass

Test Case ID	TC-07
Feature / functionality	Recommendations: Kind filter
Test type	Positive
Test input	In “What to do”, click the “Hardware upgrade” filter.
Expected result	Only Hardware-upgrade cards (R2, R3) are shown; Inefficiency and Rebalance cards are hidden.
Actual result	Filter 'Hardware upgrade' shows cards ['R2', 'R3']; expected ['R2', 'R3']; Inefficiency cards hidden
Pass / Fail	**PASS**
Justification	The filter must hide exactly the right cards. If it hides too many or too few, the operator misses actions or sees ones they filtered out.



Kind filter = Hardware upgrade

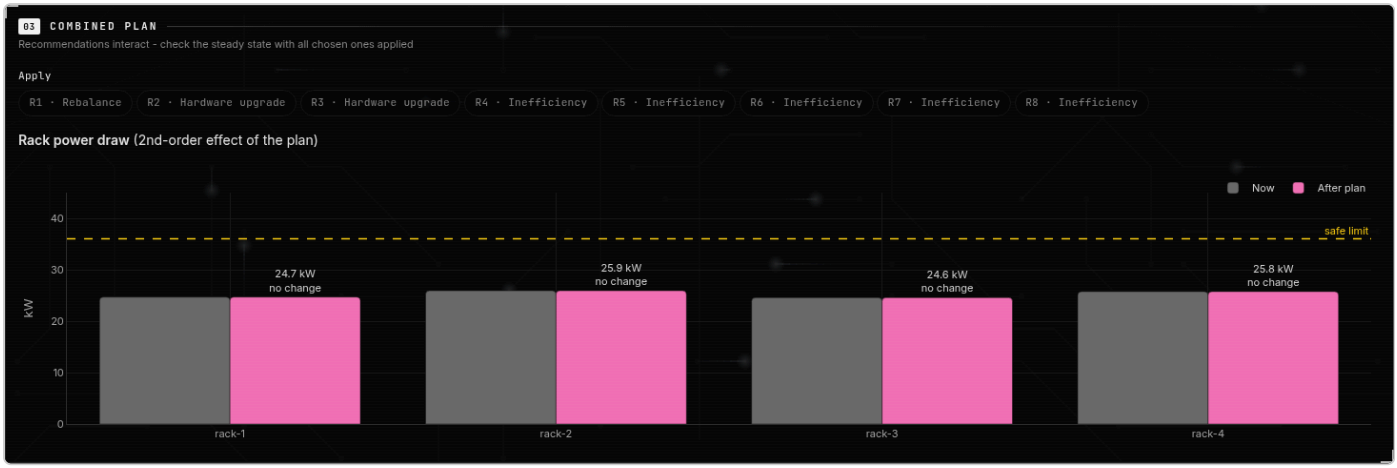
TC-08 Recommendations: Combined plan

Negative / boundary | Pass

Test Case ID	TC-08
Feature / functionality	Recommendations: Combined plan
Test type	Negative / boundary
Test input	In “Combined plan”, deselect every action pill (empty selection).
Expected result	Combined plan is empty (0 rows): the power chart shows every rack as “no change”, there are no “ends at … safe limit” warnings and no exception.
Actual result	All pills deselected: combined plan rows = 0, no 'ends at' warnings, no exception; per-card effect charts still shown (9)
Pass / Fail	**PASS**
Justification	Empty input is the classic boundary case for a multi-select. The plan maths must handle zero actions without dividing by zero or showing leftover warnings.



Combined plan with every action applied

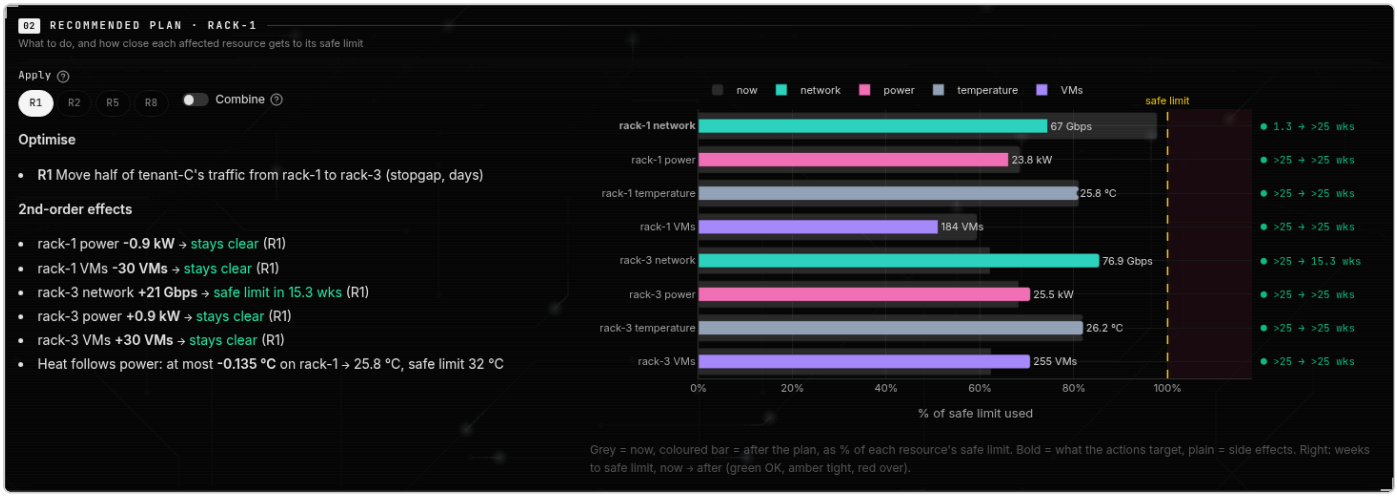


Combined plan with no action selected

TC-09 Overview: Recommended plan (Combine toggle)

Validation (required input) | Pass

Test Case ID	TC-09
Feature / functionality	Overview: Recommended plan (Combine toggle)
Test type	Validation (required input)
Test input	On the Overview tab, turn “Combine” ON, then deselect every action pill.
Expected result	The plan bar chart is not drawn and the app shows the prompt “Pick at least one action.”
Actual result	Combine ON, every pill deselected on 'ov_plan_rack-1_multi': caption shown = Pick at least one action.; no exception
Pass / Fail	**PASS**
Justification	This is the app’s version of a required-field check. The user gets told what is missing instead of a blank or broken panel.



Overview: recommended plan for the rack



Combine ON with no action selected: “Pick at least one action.”

TC-10 Recommendations: combination maths

Validation | Pass

Test Case ID	TC-10
Feature / functionality	Recommendations: combination maths
Test type	Validation
Test input	Apply R1 and R2 together; compare each shared rack × resource row with R1 alone and R2 alone.
Expected result	For every shared row: combined after = now + (R1 delta) + (R2 delta).
Actual result	R1+R2 share 3 rows; combined 'after' equals before + sum of deltas on every row (mismatches: none)
Pass / Fail	**PASS**
Justification	Actions interact, for example when two of them change the same rack's power. This checks that the combined numbers are added up correctly and nothing is counted twice.

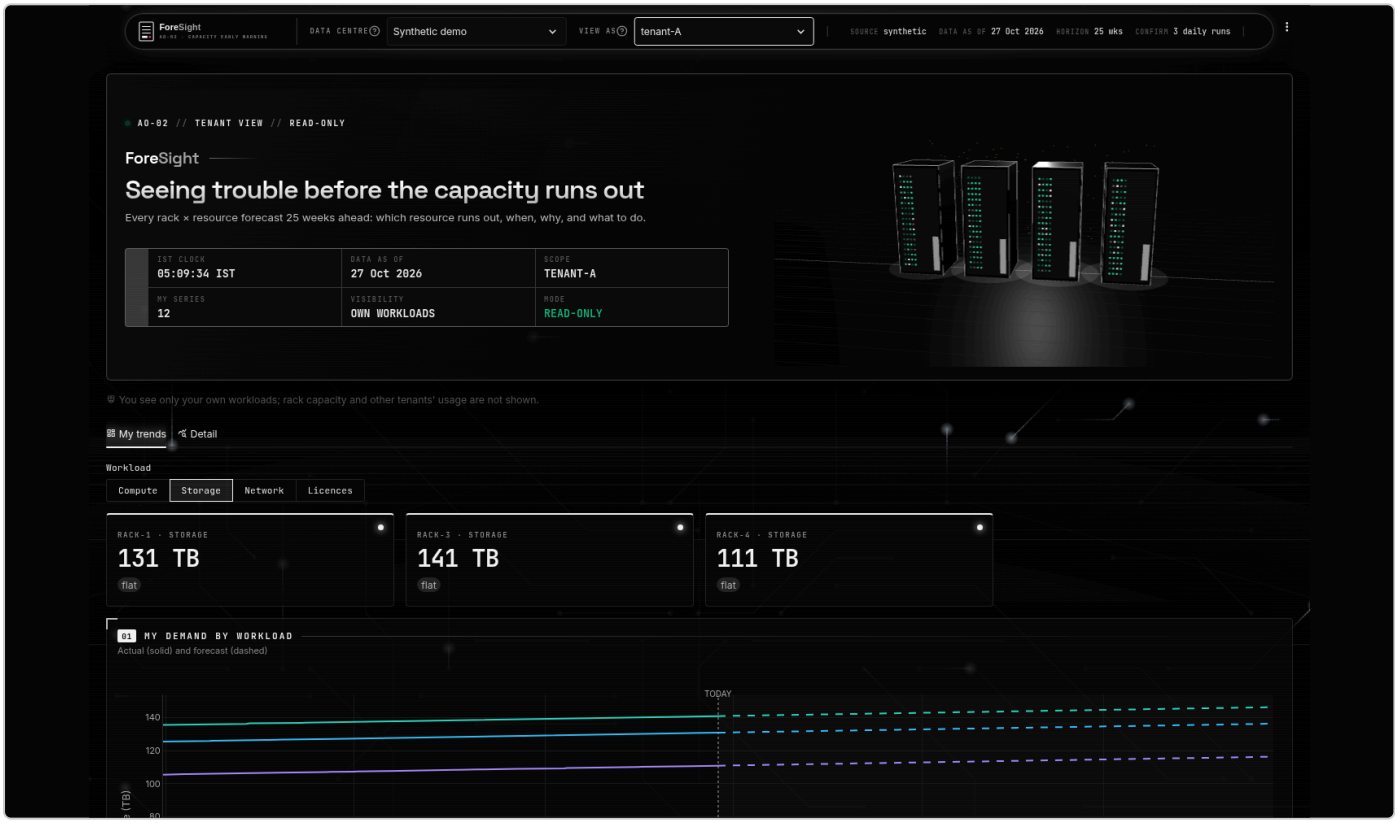


Combined plan with every action applied

TC-11 Tenant view (access control)

Validation / security | Pass

Test Case ID	TC-11
Feature / functionality	Tenant view (access control)
Test type	Validation / security
Test input	Change “View as” from Operator to “tenant-A”.
Expected result	Only “My trends” and “Detail” tabs; no Recommendations/Validation/Audit tabs; no other tenant’s name appears anywhere on the page.
Actual result	Viewing as 'tenant-A': tabs = ['My trends', 'Detail']; other tenants mentioned = none; no Recommendations/Audit tab
Pass / Fail	**PASS**
Justification	Tenant isolation is a hard requirement. A tenant must never see other tenants’ usage or rack totals.



Viewed as tenant-A: own workloads only

TC-12 Data loader: required columns

Negative | Pass

Test Case ID	TC-12
Feature / functionality	Data loader: required columns
Test type	Negative
Test input	Load a telemetry file whose time column is named “time” instead of “timestamp”.
Expected result	Load stops with a clear error naming the missing column and listing the file’s real columns (so config.py is a one-line fix).
Actual result	KeyError raised: "telemetry *telemetry*: column 'timestamp' (for 'timestamp') not found. File columns: ['time', 'rack', 'tenant', 'resource', 'value'] - fix config.py"
Pass / Fail	**PASS**
Justification	Real data comes in many layouts. A clear error is better than silently loading wrong data, and it tells the user exactly what to fix.

```
$ venv/bin/python run_tests.py   # TC-12
TC-12: PASS - KeyError raised: "telemetry *telemetry*: column 'timestamp' (for 'timestamp') not
found. File columns: ['time', 'rack', 'tenant', 'resource', 'value'] - fix config.py"
```

TC-13 Data loader: invalid values

Negative / validation | Pass

Test Case ID	TC-13
Feature / functionality	Data loader: invalid values
Test type	Negative / validation
Test input	Telemetry values “5”, “abc”, “6”, “” (empty); limits row with rack “*”.
Expected result	Non-numeric and empty values are dropped (2 rows kept: 5, 6); the “*” limit is expanded to every rack (r1).
Actual result	4 input rows → 2 kept, values [5.0, 6.0] ('abc' and empty dropped); wildcard limit '*' expanded to racks ['r1']
Pass / Fail	**PASS**
Justification	Real telemetry has gaps and junk values. They must be dropped, not passed to the forecast as zeros or crash it.

```
$ venv/bin/python run_tests.py   # TC-13
TC-13: PASS - 4 input rows → 2 kept, values [5.0, 6.0] ('abc' and empty dropped); wildcard limit
'*' expanded to racks ['r1']
```

TC-14 Forecast ETA (exhaustion)

Boundary | Pass

Test Case ID	TC-14
Feature / functionality	Forecast ETA (exhaustion)
Test type	Boundary
Test input	Forecast band [70, 80, 90, 100] against thresholds 90 (hit exactly), 100.01 (never), 70 (first step); last level 95 vs threshold 90; no forecast.
Expected result	Day 3; None; Day 1; already past → 0; no forecast → None.
Actual result	threshold hit exactly → day 3; never reached → None; reached at first step → day 1; already past → likely 0; no forecast → None
Pass / Fail	**PASS**
Justification	Off-by-one errors in the ETA would show the wrong “weeks to safe limit” on every alert. This checks the edge values exactly.

```
$ venv/bin/python run_tests.py   # TC-14
TC-14: PASS - threshold hit exactly → day 3; never reached → None; reached at first step → day 1;
already past → likely 0; no forecast → None
```

TC-15 Data loader: invalid timestamp

Negative | Fail

Test Case ID	TC-15
Feature / functionality	Data loader: invalid timestamp
Test type	Negative
Test input	Telemetry with timestamps “2024-01-01”, “not-a-date”, “2024-01-03”.
Expected result	The unparseable row is dropped (2 rows kept), the same way invalid values are dropped in TC-13.
Actual result	Loader crashed with ValueError: time data "not-a-date" doesn't match format "%Y-%m-%d". You might want to try:
Pass / Fail	**FAIL**
Justification	Malformed dates are common in exported logs. One bad row should not stop the whole data centre from loading.

```
$ venv/bin/python run_tests.py   # TC-15
TC-15: FAIL - Loader crashed with ValueError: time data "not-a-date" doesn't match format "%Y-%m-%d". You might want to try:
```